

APIx: Airfare Price Index Platform (v2)

Dual-Interface Real-Time Airfare Tracking, Mathematical Laspeyres Index Engine & Regulatory Monitoring

Date: September 06, 2026 | Version: 2.0.0 Production | Target Audience: Indian Consumers, NSO, MoSPI, RBI, DGCA

1. Executive Summary & Problem Statement

APIx is an airfare price intelligence ecosystem built to solve two contrasting needs across India's domestic aviation sector:

- **Consumer Price Transparency:** Indian domestic airfares are volatile due to dynamic pricing algorithms. Everyday consumers struggle to evaluate whether today's quoted price is fair or inflated.
- **Macro-Economic & Regulatory Tracking:** Official government Consumer Price Index (CPI) metrics from MoSPI suffer from a 45-day reporting lag. Macroeconomic regulators (NSO, RBI, DGCA) require real-time, mathematically rigorous Laspeyres index tracking with daily sector weight attributions.

The Readability Rule: If a non-technical consumer cannot understand a screen in 10 seconds, the platform simplifies it into plain English (e.g., *'Flights are 10% pricier than usual today'*) while maintaining deep technical analysis behind an analyst toggle.

2. System Architecture & Tech Stack

Layer	Technologies	Key Responsibilities
Frontend UI	React 18, TypeScript, Vite, TailwindCSS, Recharts, Lucide Icons	Dual-view UI (Consumer vs Advanced Analyst), real-time chart rendering, interactive guided tour, explainers.
Backend API	FastAPI (Python 3.9+), Uvicorn, Pydantic, WebSockets	High-performance REST API, asynchronous WebSockets, JWT authentication, rate limiting, and NLP chatbot engine.
Data Pipelines	NumPy, Pandas, APScheduler, Playwright Scrapers	IQR outlier cleaning, base-period pricing, Laspeyres index calculations, and automated scraping simulation.
Persistence & Cache	SQLite 3 / PostgreSQL 15, SQLAlchemy 2.0, Redis 7	ACID transaction storage for raw quotes, daily aggregates, index history, user tiers, and fast caching.

3. Dual-Interface System Design

Feature Area	Consumer Public View (Default)	Advanced Analyst / Regulator View
Hero Metric	Large Flight Price Score (e.g. 110.3) with color-coded verdict banner.	Laspeyres Index value with mathematical formulas, DoD change, and base-period reference.
Data Transparency	[LIVE] badge with data source and human-readable timestamp.	Scraper pipeline logs, latency monitors, success rate %, and health status.
Route Analysis	Cheapest routes, Fare/km ranking, and 'Book Now' deep links.	DGCA traffic-weighted route breakdown, fare IQR spread, and raw quote drilldown.
Onboarding & Help	4-step guided walkthrough tour & 'Explain this' popups.	Tiered API key management (Bronze/Silver/Gold) & raw JSON/CSV export playground.

4. The 8 Standout Innovation Features

1. Real Traveller vs Official CPI Comparison

Dual-line tracking showing daily APIx live scores alongside official monthly MoSPI CPI data (45-day lag) and DGCA benchmark averages, validating real-time predictive inflation accuracy.

2. 'Best Time to Book' Predictor

Advance purchase lead-time curve (T+1 to T+45) highlighting the optimal 21-28 day booking sweet spot where average fares drop by 22% compared to last-minute bookings.

3. Fare Alert & Price Watch

Custom route monitoring allowing users to set target price thresholds and receive instant visual alerts when fares dip below their budget.

4. Route Affordability (Fare per KM)

Normalizes domestic routes by sector distance (km), calculating true INR/km affordability to expose which sectors are overpriced relative to flight duration.

5. Festival & Holiday Surge Overlay

Interactive chronological overlays for major Indian festivals (Diwali, Holi, Eid, New Year, IPL) explaining historical 30-50% demand-driven price surges.

6. Anomaly & Surge Root-Cause Explanations

Automated natural language explainers analyzing sudden price spikes (e.g. weather disruptions, fuel price adjustments, or high festive bookings).

7. Natural Language Search & Chatbot

Plain-English conversational query box and in-site chatbot that understands queries like 'How much did Delhi to Mumbai flights cost last week?'.

8. 'Book This Fare' Deep-Link Redirect

Direct action links allowing users to seamlessly transition from analytics to booking on MakeMyTrip / Airline portals with pre-filled dates and city pairs.

5. Mathematical Foundation & Cleaning Pipeline

A. Interquartile Range (IQR) Outlier Filter

To eliminate scraping artifacts and erroneous fare spikes, APIx computes IQR bounds per route and departure bucket:

$$IQR = Q3 - Q1 \quad | \quad \text{Valid Range} = [Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$$

Quotes falling outside this window are flagged as outliers and excluded from index aggregation.

B. Laspeyres Price Index Formula

The composite Airfare Price Index is calculated using the standard macroeconomic Laspeyres formulation:

$$I_t = \left[\frac{\sum (P_{i,t} \times W_{i,0})}{\sum (P_{i,0} \times W_{i,0})} \right] \times 100$$

Where $P_{i,t}$ is the median fare for route i on day t , $P_{i,0}$ is the base period price, and $W_{i,0}$ is the DGCA passenger traffic volume weight for route i .

6. Database Schema & Data Models

Table Name	Primary Fields	Description & Role
routes	id, origin_code, origin_city, destination_code, destination_city, dgca_traffic_weight, distance_km	Domestic city pairs and their official regulatory traffic weight allocations.
carriers	id, code (6E, AI, SG, QP, IX), name, is_active	Airlines tracked across domestic sectors.
raw_fare_quotes	id, route_id, carrier_id, source_id, fare_amount, base_fare, taxes_fees, flight_date, scrape_timestamp, is_outlier	Granular ticket quotes scraped or simulated across advance purchase windows.
daily_route_aggregates	id, route_id, date, median_fare, mean_fare, min_fare, max_fare, std_dev, sample_count	Cleaned daily summary statistics computed post-IQR filtering.
apix_index_values	id, date, index_value, dod_change_pct, mom_change_pct, base_period_date	Final macro-level composite daily price index records.
users	id, username, hashed_password, role, api_key, api_tier, api_limit, is_active	Role-based access control (Admin, Analyst, NSO/RBI API clients).

7. REST API Endpoints Reference

Method	Endpoint	Description & Access Tier
GET	/api/v1/index/current	Latest headline score, verdict string, and DoD change.
GET	/api/v1/index/timeseries	Historical index values (7d, 30d, 90d, 1y).
GET	/api/v1/routes/summary	Route-level median fares, price changes, and fare/km.
GET	/api/v1/analytics/cpi-comparison	APIx real-time score vs MoSPI CPI & DGCA monthly benchmark.
GET	/api/v1/analytics/advance-purchase	Lead-time pricing curve (T+1 to T+45 days).
POST	/api/v1/chatbot/query	Natural language inquiry engine & conversational guide.
GET	/api/v1/export/csv	Bulk regulatory data export (Requires Bronze/Silver/Gold key).
WS	/ws/live-stream	WebSocket connection for live quote feeds and index updates.

8. Runtime Execution & Verification Commands

Launching Backend Server:	
cd backend && python main.py	# Runs on http://localhost:8000
Launching Frontend Client:	
cd frontend && npm run dev	# Opens on http://localhost:5173
Running Offline Test Suite:	
cd backend && python -m pytest	# Validates index math, IQR rules & API routes